

Removing the `min_pool_cost` parameter

INTERNAL IOHK REPORT — NOT FOR PUBLIC DISTRIBUTION

Aikaterini-Panagiota Stouka Lars Brünjes
Aggelos Kiayias Elias Koutsoupias

March 22, 2022

Background and Proposed Action

Each user who registers a pool declares their pledge, margin, and cost of the pool. The cost reflects the amount of ADA that they will need to run the pool for one epoch. This cost will be removed from the pool's rewards and will be paid to the pool operator to compensate them for the cost they incurred to run the pool. Since the Shelley launch, there has been a restriction (called *minimum cost*, `min_pool_cost` or “mincost” for short) on the lowest amount that the pool operators can declare as cost. This minimum cost is set to 340 ADA.

Our proposal is to completely remove the minimum cost. The alternative, which is to keep it, but reduce the current minimum cost to a reasonable value, is inferior due to the fact that it should be continuously monitored and updated. Lacking an on-chain oracle of current ADA price this can only be accomplished by a continuous governance procedure.

The role of cost declarations on the Cardano system

In this subsection we explain the role of declaring the pool's cost on the Cardano System.

- The pool operators who create a *successful pool* (a pool whose rewards are higher than its cost) are able to get compensated for the cost they incurred.
- By allowing the operators to declare their cost, it is possible to improve the efficiency of the system — without cost declarations, becoming a saturated pool at equilibrium can only depend on the level of pledging.

The role of the minimum cost on the Cardano system

Since Shelley was launched there has been a minimum amount of ADA that the pool operators can declare as cost which is 340 ADA. This lower bound on cost was imposed, because there was a fear that in the beginning the system would not be “mature” enough and as a result the pool operators would declare

zero cost in order to become competitive (“race to the bottom”), although this choice may not be rational. Note that in the research paper ¹ we have proved that *at the equilibrium the players do not have any incentive to lie about their cost*. We note that cost declarations should be updated to reflect fluctuations of Ada.²

One additional reason for imposing the lower bound on cost was that it increases the amount of stake (*Sybil bound*) that an attacker needs in order to attack the system by creating multi pools and declaring cost lower than their real cost. This scenario is examined on Theorem 5 of the arxiv version of the research paper.³

Why the current minimum cost should be removed

Although 340 ADA was an indicative value for the cost of a pool for one epoch when the price of ADA was 0.10\$, now the price of ADA has increased significantly and thus 340 ADA is a much higher amount of money compared to the real cost a pool needs to run a pool.

This raises the following concerns:

- The pool operators are not allowed to declare their real cost. This has a higher impact on small pool operators. Note that if we subtract 340 ADA from the rewards of a small pool, the remaining rewards that will be shared among the members could be very low. Such a pool cannot attract new members. This would be improved, if minimum cost was removed and the pool operators could declare a value closer to their real cost.
- Recall that the cost is subtracted from the pool’s rewards and is added to the pool operator’s rewards. Thus some pool operators split their pledge among multiple pools just to collect 340 ADA from more pools.

The community has voiced concerns regarding the unfairness of minimum fee for small pools as also evident in two CIP proposals:

- <https://cips.cardano.org/cips/cip23/>, and
- <https://forum.cardano.org/t/prepay-min-fixed-fee-cip/81605>

Analyzing the above two proposals is beyond the scope of the current document. However we should mention the following: (1) the first proposal for adding a variable fee, amounts to amending the reward mechanism so that it increases slower up to saturation. This modification interferes with our original purpose of having cost declarations which is to directly compensate the pool owners for their actual operational costs. (2) The second proposal suggests that the system finds all the pools that produced blocks during the last epochs and it pays them the minimum cost for the next epoch. If the pools that produce blocks are too many (say, more than $N \cdot k$ where N is a small integer), then the system will pay a smaller amount than the minimum cost for each pool. The

¹L. Brünjes, A. Kiayias, E. Koutsoupias and A. Stouka, “Reward Sharing Schemes for Stake Pools,” in 2020 IEEE European Symposium on Security and Privacy (EuroS&P), Genoa, Italy, 2020 pp. 256-275.

²See “The general perspective on staking in Cardano,” <https://iohk.io/en/blog/posts/2020/11/13/the-general-perspective-on-staking-in-cardano/>.

³Refer to <https://arxiv.org/abs/1807.11218>.

drawback of this proposal is that it may incentivize the pool operators (even the small ones) to split their pools so that they receive multiple times the “free” minimum cost. Moreover, the motivation for introducing this proposal is much better addressed by simply removing the mincost parameter — see below.

Why removing the mincost will mitigate the above concerns

If we remove the minimum cost, then the above problems will be mitigated because the users will be able to declare their real cost and thus we expect that the cost the pool operators will declare will be much lower than 340 ADA. Importantly, this also prevents pool operators from splitting their pledge among multiple pools just to collect the reward from the high declared cost.

Why removing mincost will not harm the system

Note that now the system is more “mature” and we expect operators to be able to find the right balance between the declared cost and margin. That is, we expect that eventually they will truthfully declare their cost and adjust appropriately their margin to be sufficiently competitive.

Moreover, we analyze below for completeness the impact of removing minimum cost (i) when the players are rational or (ii) when there exists a malicious attacker that declares zero cost just to collect power and attack the system.

No incentives to lie about cost at equilibrium

As we have already explained above, we have proved that *at the equilibrium all the (rational) players do not have incentives to lie about their cost*, something that prevents the race to the bottom (see page 23 on the arxiv version of the research paper <https://arxiv.org/abs/1807.11218>).

The intuition behind this is that when the players declare lower cost than their real one, they have to pay on their own for the cost they hid. We provide some examples to explain this further.

Recall that we order the players based on their *potential profit* which reflects the profit of their pool if it was saturated. In addition *desirability* of a pool reflects the rewards that this pool would give to the members if it was saturated. At the equilibrium players $1, \dots, k$ open a saturated pool and set their margin accordingly so that their pools have the same desirability as the potential profit of player $k + 1$. Also, their rewards from margin will be the difference between their potential profit and the potential profit of $k + 1$.

Why players $k, k + 1, \dots, n$ do not have incentives to declare lower cost than their real one at equilibrium. Let us examine the case where the player $k + 1$ declares lower cost than its real one in order to increase their potential profit and get ranked among the best k . We will show that their utility cannot increase. We use data from the snapshot available as a spreadsheet⁴, where the $k + 1$ player has potential profit 53537 ADA. Their utility at the equilibrium will be $0.024585 \cdot 53537$ which reflects their potential profit multiplied by the ratio of their stake out of $1/k$ (which is the size of the pool to which they

⁴https://docs.google.com/spreadsheets/d/1MbAs9kuzFz_AmvsxGLTd_fdZS8IaeBPqBYDqJB1ZdZk/edit?usp=sharing

have delegated their stake). Let us assume that this player declares lower cost so that their potential profit becomes the same as player's k potential profit, and they take their position in the ranking. This means that they create a saturated pool at the equilibrium. If x is the cost they hide from their real cost (this means that $c' = c - x$ where c is their real cost and c' is the cost they declared) their utility will become $0.024585 \cdot 53537 + x \cdot (0.024585 - 1)$ which is lower.

Why players $1, \dots, k$ do not have incentives to declare lower cost than their real one at equilibrium. Let us assume that a player i ($i \leq k$) hides x ADA from cost. Then they will remain in the best k and will earn x more rewards from margin. This happens because when they hide x cost, their potential profit will increase by x and thus they will be able to set higher margin and still remain in the best k . However, they will have to pay x more ADA for cost they hid. Thus their utility will be exactly the same.

Sybil bound-computations

At this point we analyze the scenario of a malicious attacker who tries to attack the system by creating multiple pools, declaring zero cost and trying to collect relative power (collective delegated stake and pledge) that is higher than $1/2$. The lower bound on the stake the attacker needs to succeed on this attack is called *Sybil bound*.

We use some values from a snapshot of Cardano to compute the Sybil bound with and without the minimum cost, and we demonstrate that the difference is not very high. In the following paragraphs we describe these computations.

In order for a malicious attacker to obtain control of $1/2$ of stake or more at an equilibrium, they should have (relative stake) at least $S = k/2 \cdot [\lambda_{k/2+1} - \frac{c_{k/2+1} - c^*}{R} \cdot (1 + \frac{1}{a_0})]$ where c^* is the cost they declare and $\lambda_{k/2+1}, c_{k/2+1}$ are the relative pledge and the cost of the player with *potential profit order* $k/2 + 1$ (1 has the player with the highest potential profit).

Lower bound of the attacker who declares the min cost Assuming that the min cost is 340 ADA, $c^* = 340$, $k = 500$, $\lambda_{251} = 0.00001092808928$, $\alpha_0 = 0.3$, $R = 35000000$, $c_{k/2+1} = 340$ (the snapshot used is available as a spreadsheet⁵) we have $S = 0.00273202232$. Note that S is the relative stake, in absolute terms this is approximately $8.5 \cdot 10^7$ ADA.

Lower bound of the attacker who declares zero cost If we remove min cost, we consider $c^* = 0$. If we consider that $c_{k/2+1} = 30$ which is an indicative value for the real cost of a pool then $S = 0.00180345089$. This in absolute terms is approximately $5.61 \cdot 10^7$ ADA.

⁵https://docs.google.com/spreadsheets/d/1MbAs9kuzFz_AmnbxGLTd_fdZS8IaeBPqBYDqJB1ZdZk/edit?usp=sharing